

Obstructive sleep apnoea and obesity.

PMID: 28697288 | Indexed in PubMed

Obstructive sleep apnoea (OSA) and obesity are two common conditions affecting the Australian population; obesity is the major risk factor for OSA. The objectives of this article are to review the interactions between obesity and OSA, including the increased cardiovascular risk, and highlight the importance of using OSA diagnosis as a critical time to address obesity itself and other cardiovascular risk factors. Snoring and symptoms of OSA frequently worsen during periods of rapid weight gain. Obesity and metabolic factors (eg hypertension, hyperlipidaemia, insulin resistance) are commonly present at the time of OSA diagnosis. Severe OSA is an independent risk factor for cardiovascular disease and stroke. Continuous positive airway pressure (CPAP) treatment is safe and effective, reduces sleepiness, and improves mood and quality of life; however, the cardiovascular benefits of CPAP are uncertain. Weight loss leads to variable improvement in OSA, is most effective in milder OSA, and has a greater benefit on cardiovascular risk than CPAP.

Obstructive sleep apnea and cardiovascular disease - time to act!

PMID: 17525876 | Indexed in PubMed

Sleep related breathing disorders are common and their potential to disrupt sleep leading to daytime fatigue and hypersomnolence is widely acknowledged. In the future, obstructive sleep apnoea (OSA) may become even more important because obesity as a main risk factor is increasingly prevalent. Apart from disturbing sleep, OSA has also been recognised as a risk factor for hypertension, acute cardiovascular events and metabolic disturbance such as insulin resistance. Several randomised controlled trials demonstrated a positive effect of nasal continuous positive airway pressure (CPAP) treatment on arterial blood pressure, leading the "Joint National Council on High Blood Pressure" to list obstructive sleep apnoea as the first identifiable cause of arterial hypertension. Recently, a growing body of evidence demonstrated also a risk reduction of fatal and non-fatal cardiovascular events by treatment of obstructive sleep apnoea. A beneficial effect of treatment of OSA was also shown for patients with heart failure, or heart rhythm disturbance. Obstructive sleep apnoea may no longer be seen as a cause for daytime sleepiness and impaired quality of life only, but also as an independent risk factor, at least for the occurrence of hypertension but probably for any cardiovascular and cerebrovascular disease. While prospective controlled trials to document a reduction of cardiovascular morbidity and mortality are awaited, therapeutic nihilism seems no longer appropriate. With effective treatment available, subgroups that may profit best remain to be identified.

[Obstructive sleep apnea and risk factors for coronary artery disease].

PMID: 11004862 | Indexed in PubMed

Obstructive sleep apnoea (OSA) is strongly associated with obesity, especially abdominal obesity. Obesity in turn is a well-known risk factor for coronary artery disease (CAD). The aim of our study was to evaluate the relationship between OSA severity and CAD risk factors. The sample consisted of 73 subjects (mean age $\pm$ SE, 46.7 $\pm$ 1 years) referred to a sleep laboratory. Subjects were either: 1. obese with OSA (O-OOSA group n = 35; body mass index, BMI $\geq$ 30 kg/m²; apnoea/hypopnoea index, AHI $>$ 35), 2.

non-obese with OSA (BO-OSA group $n = 14$; BMI 35), or 3. obese without OSA (O-Z group $n = 24$; BMI ≥ 30 kg/m²; AHI < 5). All subjects underwent full overnight polysomnography. Blood samples were taken from all subject, for fasting levels of insulin (INS), glucose (GLU), total, HDL and LDL cholesterol, triglyceride (TG) and uric acid (UA). O-OSA had significantly higher INS and UA levels ($p < 0.05$) compared to BO-OSA and O-Z. GLU and lipid levels were comparable in the studied groups. GLU level correlated ($p < 0.05$) negatively to minimum oxyhemoglobin saturation (SAT-MIN) and positively to neck circumference. TG and UA levels were correlated ($p < 0.05$) positively to AHI and negatively to SAT-MIN. UA level was also positively correlated ($p < 0.05$) to BMI, waist/hip circumference ratio (WHR), and INS level. INS level correlated ($p < 0.05$) positively to AHI, T90, WHR and UA, and negatively to SAT-MIN and mean oxyhemoglobin saturation. After adjusting for the influence of OSA and obesity (multiple regression analysis), we found independent negative correlations ($p < 0.05$) between: GLU level and SAT-MIN, UA level and SAT-MIN, and INS level and SAT-MIN. An independent, positive correlation ($p < 0.05$) was found between TG level and AHI. Results of our study suggest that OSA increases the risk of coronary artery disease by increasing plasma levels of glucose, triglyceride and insulin, independent of obesity.

Weight loss as a treatment for obstructive sleep apnoea.

PMID: 17210276 | Indexed in PubMed

Obesity is a major health problem in well-developed societies. It is frequently associated with metabolic, cardiovascular and psychological comorbid conditions. Besides, it is known that obesity constitutes a risk factor for obstructive sleep apnoea (OSA). Weight loss has been evaluated as a treatment for OSA. We reviewed the nature of the link between obesity and OSA as well as the conservative and surgical therapeutic approaches of obesity. We consider the effectiveness of weight loss alone or combined with other modalities of treatment in OSA. We conclude that although weight loss is important and can facilitate the treatment of OSA, it can rarely cure it without being associated to classical techniques, such as nasal continuous positive airway pressure (nCPAP).

Sleep studies in obese children with obstructive sleep apnea: Pulse oximetry as a diagnostic tool.

PMID: 38634170 | Indexed in PubMed

Obstructive sleep apnea is associated with many co-morbidities in children and young people. Obesity has long been recognised as an important risk factor associated with obstructive sleep apnea. Currently, polysomnography is regarded as the gold-standard diagnostic tool for obstructive sleep apnea. The validity of oximetry as a screening and diagnostic tool for obstructive sleep apnea has been the subject of several studies in the literature. There is little published evidence on the use of oximetry in children with obesity. This study assesses whether oximetry is a reliable diagnostic tool for obstructive sleep apnea in obese children. We reviewed the medical records of obese children with a suspected diagnosis of obstructive sleep apnea who underwent oximetry and cardiorespiratory polygraphy or full polysomnography at Sheffield Children's Hospital between January 2010 and March 2022. We compared oximetry results with the apnea-hypopnea index from cardiorespiratory polygraphy or full polysomnography for each patient. A total of 60 patient records were included in the analysis. The sensitivity of oximetry in diagnosing obstructive sleep apnea was 70.9%, with a specificity of 65.5% and a

positive predictive value of 68.75%. In the subgroup of subjects with severe obstructive sleep apnea (apnea-hypopnea index > 10), oximetry had a sensitivity of 87.5%. We concluded that oximetry could be a helpful initial diagnostic tool for obstructive sleep apnea in obese children, but is not entirely reliable. A negative oximetry result in a symptomatic individual should prompt a referral for more detailed investigations, while a positive result can help in treatment decisions without needing a polysomnography.

Association Between the Weight-Adjusted Waist Index and OSA Risk: Insights from the NHANES 2017-2020 and Mendelian Randomization Analyses.

PMID: 39583933 | Indexed in PubMed

Obesity is a significant risk factor for obstructive sleep apnea (OSA). The weight-adjusted-waist index (WWI) reflects weight-independent centripetal obesity. Our study aims to evaluate the relationship between WWI and OSA. The data used in the current cross-sectional investigation are from the National Health and Nutrition Examination Survey (NHANES), which was carried out between 2017 and 2020. We utilized weighted multivariable-adjusted logistic regression to evaluate the relationship between WWI and the risk of OSA. In addition, we applied various analytical methods, including subgroup analysis, smoothing curve fitting, threshold effect analysis and the receiver operating characteristic (ROC) curve. To further explore the relationship, we conducted a MR study using genome-wide association study (GWAS) summary statistics. We performed the main inverse variance weighting (IVW) method along with other supplementary MR methods. In addition, a meta-analysis was conducted to provide an overall evaluation. WWI was positively related to OSA with the full adjustment [odds ratio (OR)=1.14, 95% confidence interval (95% CI): 1.06-1.23, $P<0.001$]. After converting WWI to a categorical variable by quartiles (Q1-Q4), compared to Q1 the highest WWI quartile was linked to an obviously increased likelihood of OSA (OR=1.26, 95% CI: 1.06-1.50, $P=0.01$). Subgroup analysis revealed the stability of the independent positive relationship between WWI and OSA. Smoothing curve fitting identified a saturation effect of WWI and OSA, with an inflection point of 11.62. In addition, WWI had the strongest prediction for OSA (AUC=0.745). Sensitivity analysis was performed to verify the significantly positive connection between WWI and stricter OSA (OR=1.18, 95% CI: 1.05-1.32, $P=0.005$). MR meta-analysis further supported our results (OR=2.11, 95% CI: 1.94-2.30, $P<0.001$). Sensitivity analysis confirmed the robustness and reliability of these findings. WWI was significantly associated with the risk of OSA, suggesting that WWI could potentially serve as a predictor for OSA.

A comparative analysis of anthropometric indices for diagnosing obstructive sleep apnea among American Adults.

PMID: 39609575 | Indexed in PubMed

The occurrence of obstructive sleep apnea (OSA) syndrome is related to not only the overall obesity, but also the local distribution of fat. Therefore, we introduced anthropometric indicators to comprehensively assess the risks associated with obesity, including body mass index (BMI), which is used to assess the overall obesity, together with waist circumference (WC), body roundness index (BRI), visceral adiposity index (VAI) and lipid accumulation product (LAP), all of which reflect fat distribution. This cross-sectional study analyzed data from the 2005-2008 and 2015-2018 NHANES cycles, comprising 8966 participants.

Logistic regression models were used to examine the associations between anthropometric indices and OSA, adjusting for relevant covariates. Receiver operating characteristic (ROC) curves were constructed, and values of the area under the curve (AUC) were calculated to assess diagnostic performance. AUC comparisons between men and women were conducted using the DeLong's test. After adjusting for covariates, WC and BMI emerged as the strongest diagnostic indicators of OSA risk. A ROC analysis revealed that WC and BMI had higher diagnostic performance among women (AUC: WC: 0.651, BMI: 0.653) than men (AUC: WC: 0.616, BMI: 0.619). Other indices demonstrated weaker diagnostic abilities. Both WC and BMI demonstrated a strong diagnostic ability for OSA across both sexes, with slightly better performance observed among women. These findings suggest that WC and BMI are practical and effective tools for the early detection, risk stratification and management of OSA in clinical practices.

Pregnancy-onset obstructive sleep apnea and ambulatory blood pressure.

PMID: 39584369 | Indexed in PubMed

Elevated nocturnal blood pressure (BP) increases the risk for hypertensive disorders of pregnancy (HDP). Though obstructive sleep apnea (OSA) increases the risk for HDP, data on OSA, and 24-hour ambulatory blood pressure monitoring (ABPM) in pregnancy are scarce. We aim to examine the BP profile of women with pregnancy-onset OSA. Pregnant women with overweight/obesity and snoring were recruited at <13 weeks gestation and underwent level III home sleep apnea testing and 24-hour ABPM at enrollment and at 31-34 weeks' gestation. Women with OSA at enrollment were excluded. Mean differences in nocturnal BP and 24-hour BP measurements, between women with pregnancy-onset OSA and women without OSA, were computed by multivariable linear regression. Participants (40/101) had pregnancy-onset OSA (respiratory event index (REI) ≥ 5 events per hour) in the third trimester. Despite no significant differences in baseline BP, nocturnal systolic and diastolic BP were significantly higher in women with pregnancy-onset OSA compared to women without OSA, after adjusting for multiple covariates (mean difference 5.49 (1.45-9.52) and 3.89 (0.19-7.60), respectively). Differences in systolic BP persisted into the daytime in the OSA group. Lack of nocturnal dipping was highly prevalent in both groups, but the difference was not significant in the adjusted model. Pregnancy-onset OSA in the 3rd trimester is associated with elevated nocturnal BP and daytime systolic BP. The lack of nocturnal dipping irrespective of REI cut-off in pregnant women at risk for SDB further demonstrates the limitations of REI in defining pathology in pregnancy.

Risk factors and treatment for obstructive sleep apnea amongst obese children and adults.

PMID: 19532087 | Indexed in PubMed

The purpose of this review is to summarize the evidence for the association between obesity and obstructive sleep apnea (OSA), as well as predisposing risk factors and treatment strategies for OSA amongst obese patients. Recent findings highlight a number of factors including sex, age, upper airway structure and ethnicity, which may predispose patients to OSA when obese. Both invasive and noninvasive weight-reduction strategies also show positive signs of being an effective means to reduce or remediate OSA amongst obese adults and children. In view of recent findings, a direct association

between body mass and upper airway obstruction should be viewed with caution. Obesity may play a more significant role in the predisposition to OSA amongst particular subgroups of the population, such as adults, and those with particular craniofacial and upper airway morphology. Healthcare prioritization and requirements may be more substantial for such groups. Further, commonly used treatment methods for OSA (such as adenotonsillectomy for children and continuous positive airway pressure for adults) may be less effective for obese individuals. Weight-reduction strategies appear important for an optimal outcome, and such strategies may be more or less invasive depending on the severity of obesity, OSA or both, and other patient complications.

Cardiopulmonary disorders during sleep: diagnosis and management.

PMID: 3542725 | Indexed in PubMed

Sleep-related breathing disorders may contribute to the nocturnal peak in human mortality. Nocturnal hypoxia has been associated with serious ventricular tachyarrhythmias as well as life-threatening bradyarrhythmias. Obesity and snoring, both of which increase with age, have been identified as risk factors for sleep-related breathing disorders, as have hypertension and heart disease.
